

# Summary Data on Acid Cropland and Lime Requirements in Tanzania

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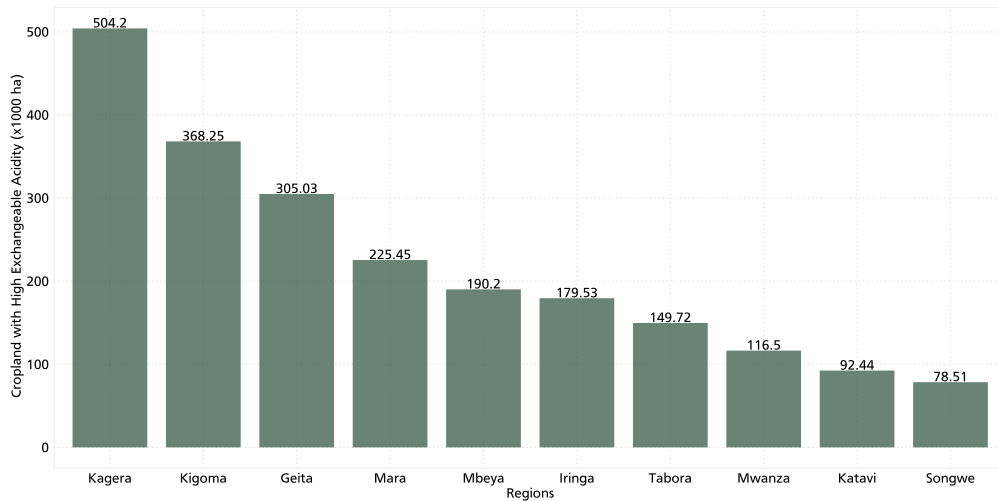
## Outlook

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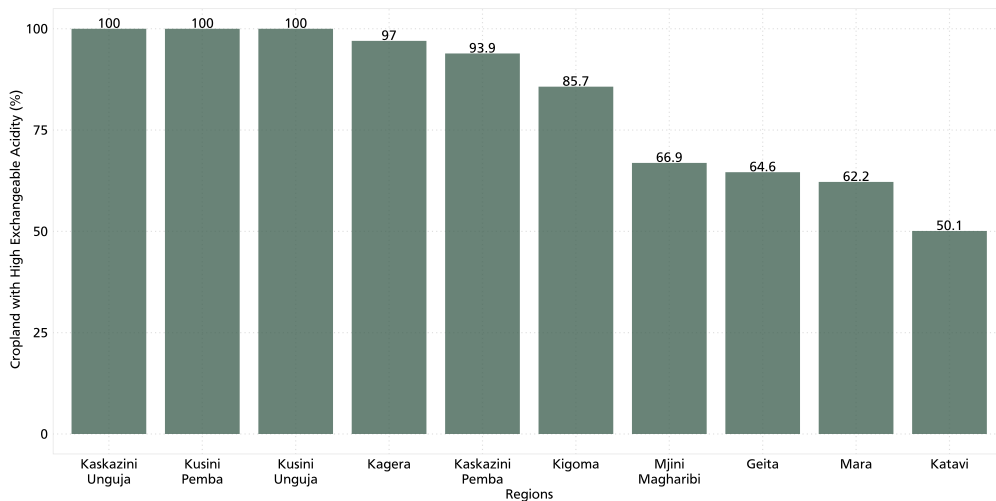
## Introduction

- The slide deck provides a **data-driven, ex-ante analysis** of soil acidity across **Tanzania**, evaluating the scale of affected areas, potential crop yield impacts, and the expected benefits of lime application for proactive soil remediation.
- **Highlights:**
  - **Acidic Cropland:** Identifies regions with high acidity levels impacting crop yields.
  - **Yield Loss:** Quantifies productivity losses in key crops due to acidic soils.
  - **Lime Requirements:** Estimates the lime amounts needed to improve soil health in these areas.
  - **Economic Potential:** Highlights estimated increases in production and production value from lime application.
- Provide policymakers, farmers, and stakeholders with actionable insights to prioritize areas with high potential for productivity gains and economic return from soil remediation.

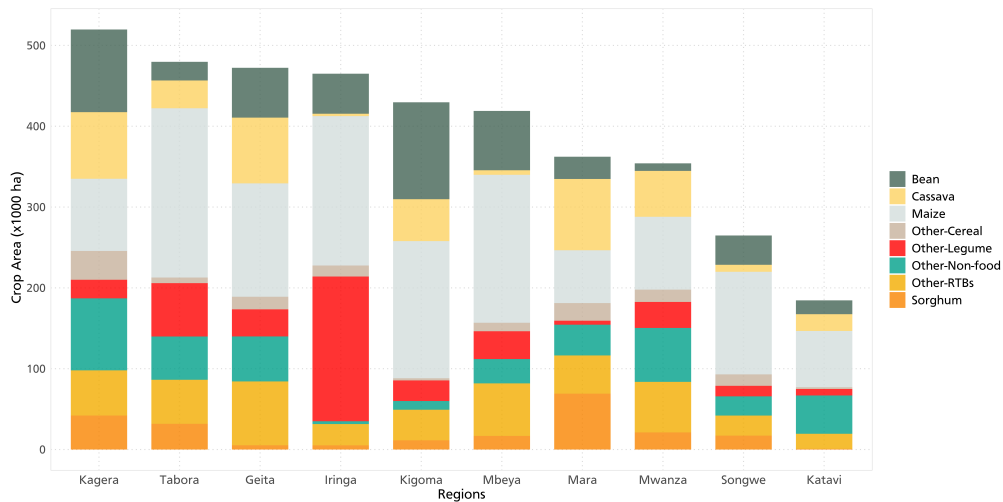
## Top 10 Regions in Tanzania with the highest acidic cropland



## Top 10 Regions in Tanzania with the highest acidic cropland (Percentage of total cropland)



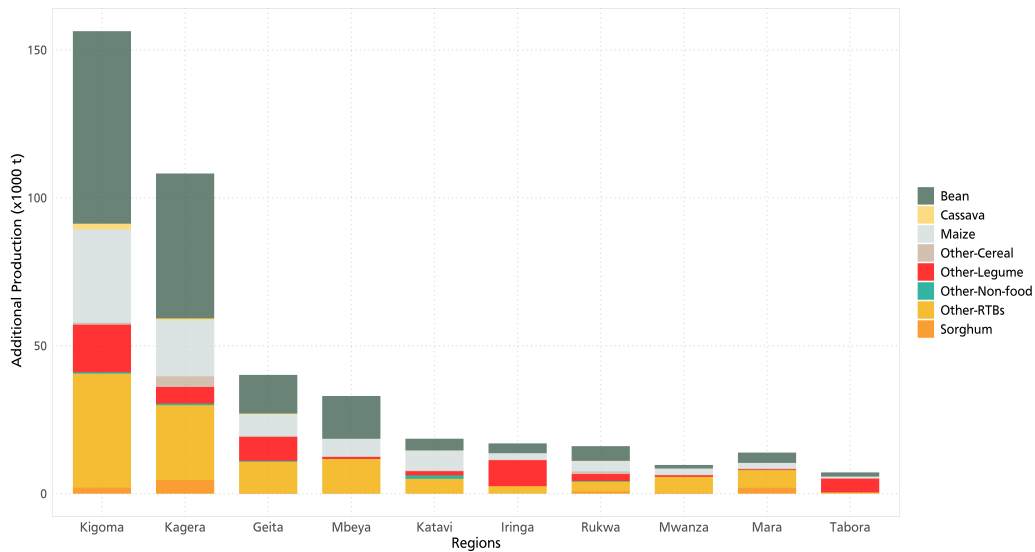
## Total crop area in the top-10 most acidic Regions, not segregated by acidic cropland



## Total crop area in the top-10 most acidic Regions, not segregated by acidic cropland (table)

| Regions | Bean   | Cassava | Maize  | Other-Cereal | Other-Legume | Other-Non-food | Other-RTBs | Sorghum |
|---------|--------|---------|--------|--------------|--------------|----------------|------------|---------|
| Geita   | 61.62  | 81.38   | 140.31 | 15.57        | 33.46        | 55.71          | 79.03      | 5.21    |
| Iringa  | 49.55  | 2.95    | 184.81 | 13.70        | 179.01       | 3.50           | 26.34      | 5.18    |
| Kagera  | 102.26 | 82.45   | 89.30  | 35.70        | 22.96        | 89.08          | 55.95      | 42.02   |
| Katavi  | 17.08  | 20.77   | 69.70  | 2.14         | 8.02         | 47.37          | 18.60      | 0.89    |
| Kigoma  | 119.90 | 51.83   | 169.76 | 2.62         | 25.52        | 10.77          | 37.86      | 11.38   |
| Mara    | 27.52  | 88.24   | 65.43  | 21.82        | 4.85         | 38.03          | 47.33      | 69.10   |
| Mbeya   | 73.37  | 5.62    | 183.11 | 10.51        | 34.35        | 30.19          | 65.01      | 16.78   |
| Mwanza  | 9.36   | 56.74   | 90.11  | 15.23        | 32.22        | 66.68          | 62.60      | 21.13   |
| Songwe  | 36.22  | 8.65    | 126.95 | 14.15        | 13.00        | 23.79          | 24.90      | 17.15   |
| Tabora  | 23.04  | 34.40   | 209.44 | 6.93         | 66.03        | 53.56          | 54.51      | 31.78   |

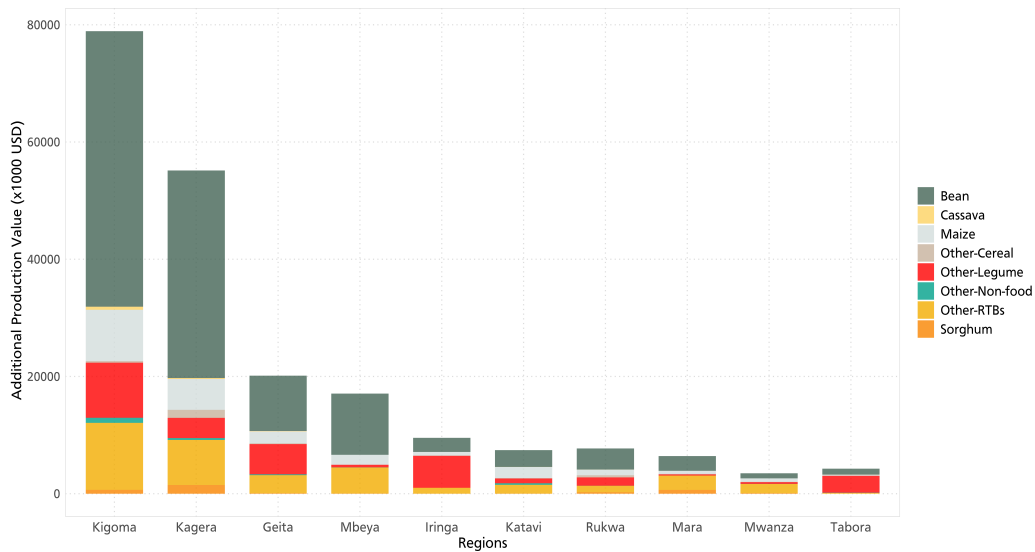
## Potential for additional production (tone) in top 10 acidic Regions in Tanzania



## Potential for additional production (1000 tone) in top 10 acidic Regions in Tanzania(Table)

| Regions | Bean  | Cassava | Maize | Other-Cereal | Other-Legume | Other-Non-food | Other-RTBs | Sorghum |
|---------|-------|---------|-------|--------------|--------------|----------------|------------|---------|
| Geita   | 13.08 | 0.25    | 7.27  | 0.37         | 8.09         | 0.27           | 10.60      | 0.22    |
| Iringa  | 3.34  | -       | 2.21  | 0.13         | 8.79         | -              | 2.52       | 0.03    |
| Kagera  | 49.01 | 0.61    | 18.95 | 3.59         | 5.61         | 0.55           | 25.32      | 4.61    |
| Katavi  | 3.97  | 0.05    | 6.77  | 0.20         | 1.39         | 1.20           | 4.94       | 0.08    |
| Kigoma  | 65.08 | 1.95    | 31.53 | 0.68         | 16.04        | 0.49           | 38.57      | 2.03    |
| Mara    | 3.47  | -       | 1.91  | 0.22         | 0.29         | -              | 6.00       | 1.99    |
| Mbeya   | 14.46 | -       | 6.06  | 0.05         | 0.71         | 0.01           | 11.61      | 0.12    |
| Mwanza  | 1.21  | -       | 2.12  | 0.16         | 0.52         | 0.01           | 5.43       | 0.27    |
| Rukwa   | 4.97  | 0.10    | 3.46  | 0.92         | 2.32         | 0.26           | 3.29       | 0.77    |
| Tabora  | 1.42  | -       | 0.67  | 0.01         | 4.66         | 0.02           | 0.41       | 0.02    |

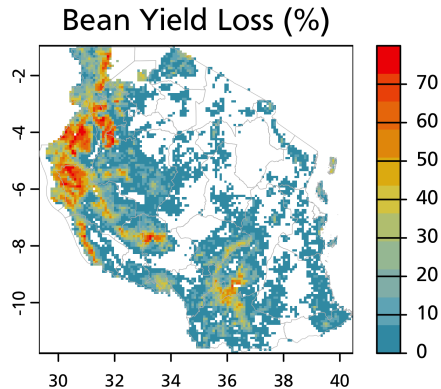
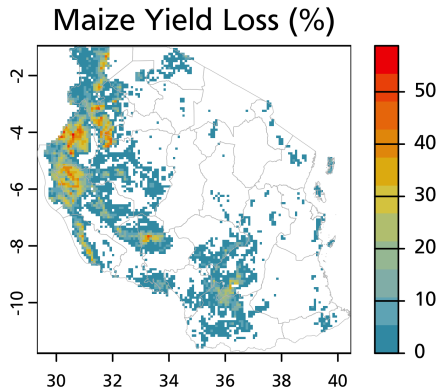
## Potential for additional production value (US\$) in top 10 acidic Regions in Tanzania



## Potential for additional production value (1000 US\$) in top 10 acidic Regions in Tanzania(Table)

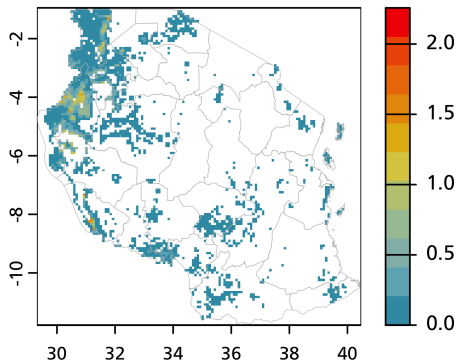
| Regions | Bean    | Cassava | Maize  | Other-Cereal | Other-Legume | Other-Non-food | Other-RTBs | Sorghum |
|---------|---------|---------|--------|--------------|--------------|----------------|------------|---------|
| Geita   | 9446.9  | 67.6    | 2019.6 | 142.1        | 5147.7       | 159.2          | 3082.8     | 71.5    |
| Iringa  | 2410.7  | 0.1     | 612.8  | 56.8         | 5444.0       | -              | 1002.3     | 8.9     |
| Kagera  | 35403.1 | 162.5   | 5264.9 | 1374.7       | 3446.0       | 327.2          | 7676.5     | 1484.5  |
| Katavi  | 2868.8  | 13.8    | 1880.5 | 75.0         | 812.6        | 265.3          | 1492.3     | 24.4    |
| Kigoma  | 47011.5 | 521.4   | 8758.5 | 258.9        | 9404.5       | 874.9          | 11426.6    | 654.4   |
| Mara    | 2509.3  | -       | 531.9  | 84.5         | 191.2        | 2.5            | 2453.4     | 642.3   |
| Mbeya   | 10445.2 | 0.8     | 1683.2 | 18.5         | 426.8        | 33.6           | 4430.3     | 39.0    |
| Mwanza  | 872.1   | -       | 589.0  | 63.1         | 295.0        | 3.4            | 1578.1     | 87.3    |
| Rukwa   | 3586.9  | 25.8    | 960.9  | 353.2        | 1424.3       | 10.1           | 1104.7     | 248.4   |
| Tabora  | 1023.3  | 1.3     | 185.6  | 5.1          | 2882.7       | 47.4           | 119.9      | 6.4     |

## Yield loss due to soil acidity for Maize and Bean in Tanzania

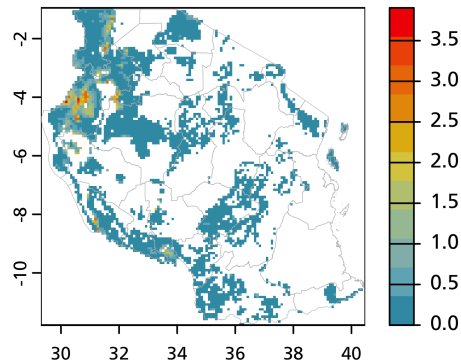


## Yield response to lime application in Tanzania

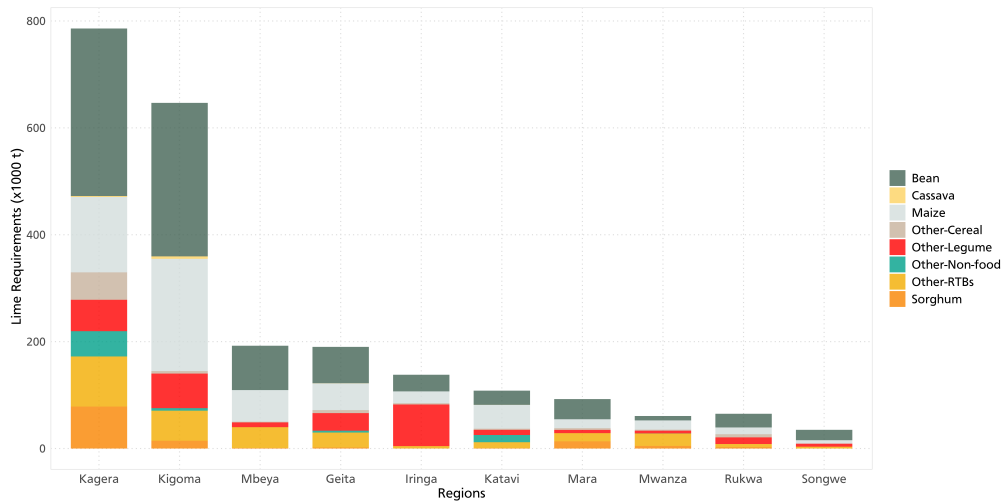
Maize Yield Response (t/ha)



Bean Yield Response (t/ha)



## Lime requirements (t) for acid soil remediation in Tanzania

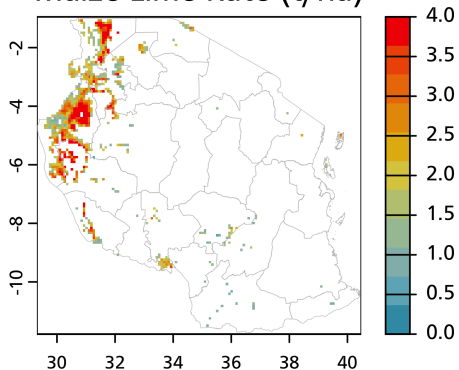


## Lime requirements (1000 t) for acid soil remediation in Tanzania (table)

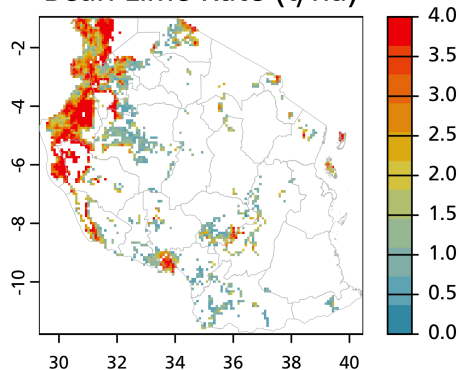
| Regions | Bean   | Cassava | Maize  | Other-Cereal | Other-Legume | Other-Non-food | Other-RTBs | Sorghum |
|---------|--------|---------|--------|--------------|--------------|----------------|------------|---------|
| Geita   | 68.13  | 0.25    | 49.96  | 5.59         | 32.92        | 3.53           | 28.08      | 1.77    |
| Iringa  | 31.07  | -       | 22.10  | 2.44         | 77.93        | -              | 4.20       | 0.27    |
| Kagera  | 313.53 | 1.77    | 141.06 | 51.32        | 58.70        | 47.33          | 93.80      | 78.43   |
| Katavi  | 26.58  | 0.14    | 44.41  | 2.19         | 9.67         | 13.49          | 11.22      | 0.56    |
| Kigoma  | 287.18 | 4.52    | 210.07 | 4.70         | 64.73        | 4.67           | 56.38      | 14.46   |
| Mara    | 37.81  | -       | 16.61  | 3.23         | 6.08         | -              | 15.38      | 13.42   |
| Mbeya   | 83.01  | -       | 59.17  | 1.32         | 8.89         | 0.02           | 38.97      | 0.96    |
| Mwanza  | 8.33   | -       | 17.33  | 1.83         | 5.06         | 0.23           | 23.33      | 4.79    |
| Rukwa   | 25.60  | -       | 12.76  | 5.83         | 12.48        | -              | 6.05       | 2.35    |
| Songwe  | 19.37  | -       | 6.32   | 0.25         | 5.53         | -              | 3.42       | 0.08    |

## Lime rate for Maize and Bean in Tanzania

Maize Lime Rate (t/ha)



Bean Lime Rate (t/ha)



## Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

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